

# What do 3, 4, 5 & 6 Sigma mean for your print & embroidery operation?

Sigma measures process quality. The higher the sigma level, the fewer defects per million opportunities, and the more reliable your output.

⚠ Most decoration businesses operate at 3 to 4 Sigma. Here's what that actually means.

**3σ**

## Industry Baseline: Where Most Start

Roughly 1 in 15 garments has a detectable defect. Misregistration, thread breaks, ink spread. Common when processes rely on operator feel over standard procedure.

**93.3%**

YIELD

66,807 DPMO

**4σ**

**Controlled: Documented Processes in Place**

Around 6 defects per 1,000 garments. Equipment calibration is routine. Operators follow standardised settings for heat, pressure, dwell time, thread tension.

**99.4%**

YIELD

6,210 DPMO

**5 $\sigma$**

### High Performance: Engineered & Measured

Only 233 defects per million. Artwork approvals locked. Stitch density standardised by fabric type. Heat press profiles stored per substrate. Statistical sampling replaces end-of-run checking.

**99.98%**

YIELD

233 DPMO

**6 $\sigma$**

### World Class: Near-Zero Defect Output

Just 3.4 defects per million. Full process capability studies per decoration type. Real-time machine monitoring. Achievable in high-volume workwear and teamwear operations with the right infrastructure.

**99.9997%**

YIELD

3.4 DPMO

#### Sigma Quality Scale



#### THE REAL GAP

Moving from **3 Sigma to 5 Sigma** cuts your defect rate by **99.6%**. That's not a machine upgrade. It's a *process redesign*. Standardised settings, documented procedures, and measured checkpoints are what separate the two. Not a newer heat press.

#### The gap is process, not machinery.

If your defect rate feels stuck, the fix usually isn't new equipment. It's documenting and standardising what your best operator already does. That's exactly what a Clarity engagement maps.

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